

FIFA World Cup 2026 Player Performance Analysis

Exploratory Data Analysis (EDA) Project Report

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Abstract

This project performs Exploratory Data Analysis (EDA) on a FIFA World Cup 2026 Player Performance dataset. The objective is to uncover patterns, trends, and relationships among player and team performance metrics. Python-based data analysis and visualization techniques were used to generate insights regarding goals, assists, ratings, positional distributions, and overall tournament performance.

Introduction

Exploratory Data Analysis is an important step in the data science workflow. It helps analysts understand the structure of a dataset, identify trends, discover anomalies, and generate meaningful insights. This project focuses on football analytics by examining player performance during the FIFA World Cup 2026.

Project Objectives

- Analyze player and team performance statistics.
- Identify top-performing players based on ratings, goals, and assists.
- Examine team-wise scoring patterns.
- Study positional distribution of players.
- Explore correlations among performance variables.

Dataset Description

The dataset contains player-level match statistics and tournament performance indicators including player ratings, goals, assists, shots, passes, defensive actions, physical metrics, and tournament ratings. Key fields include `player_name`, `team`, `position`, `goals`, `assists`, `player_rating`, `performance_score`, and `tournament_rating`.

Tools and Technologies Used

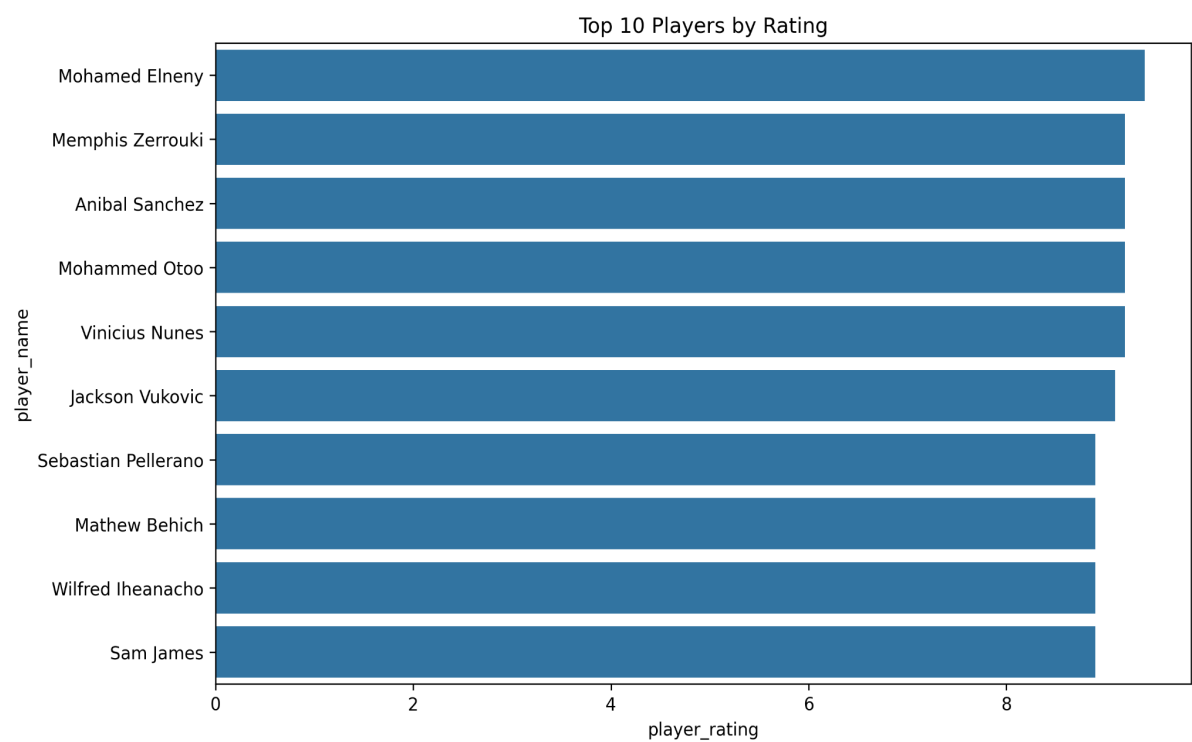
Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab, and GitHub were used to clean, analyze, visualize, and document the project.

Data Preprocessing

The dataset was imported into Google Colab and examined for structure, missing values, duplicates, and data quality issues. Numerical and categorical features were validated and prepared for analysis. Additional performance indicators such as goal efficiency were derived for deeper analysis.

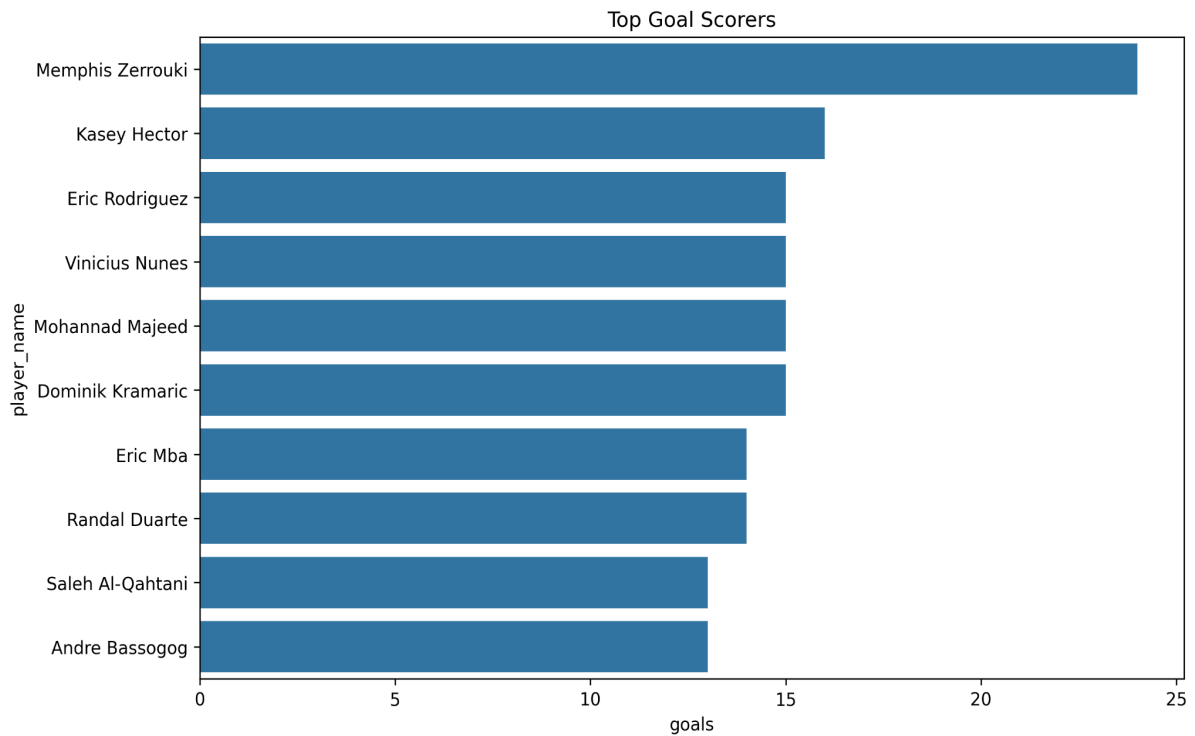
Exploratory Data Analysis and Visualizations

Top 10 Players by Rating



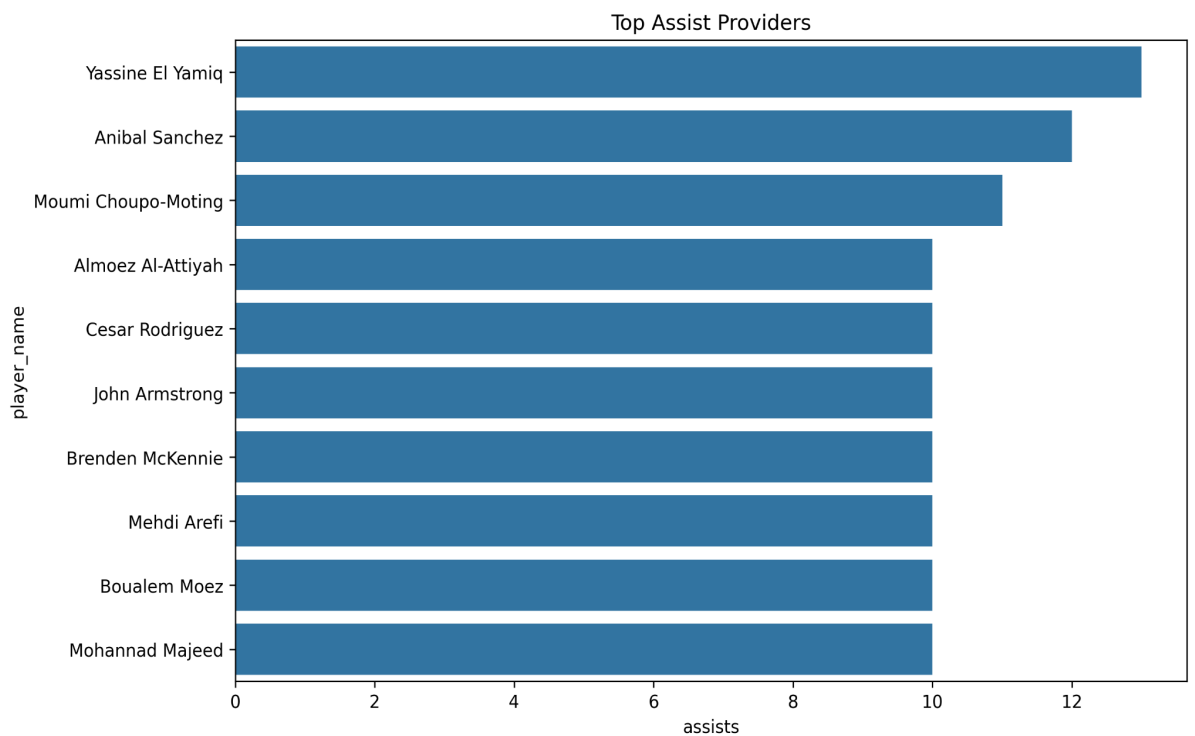
Observation: Mohamed Elneny achieved the highest player rating in the dataset. The ratings among elite performers were closely grouped, indicating strong competition among top players.

Top Goal Scorers



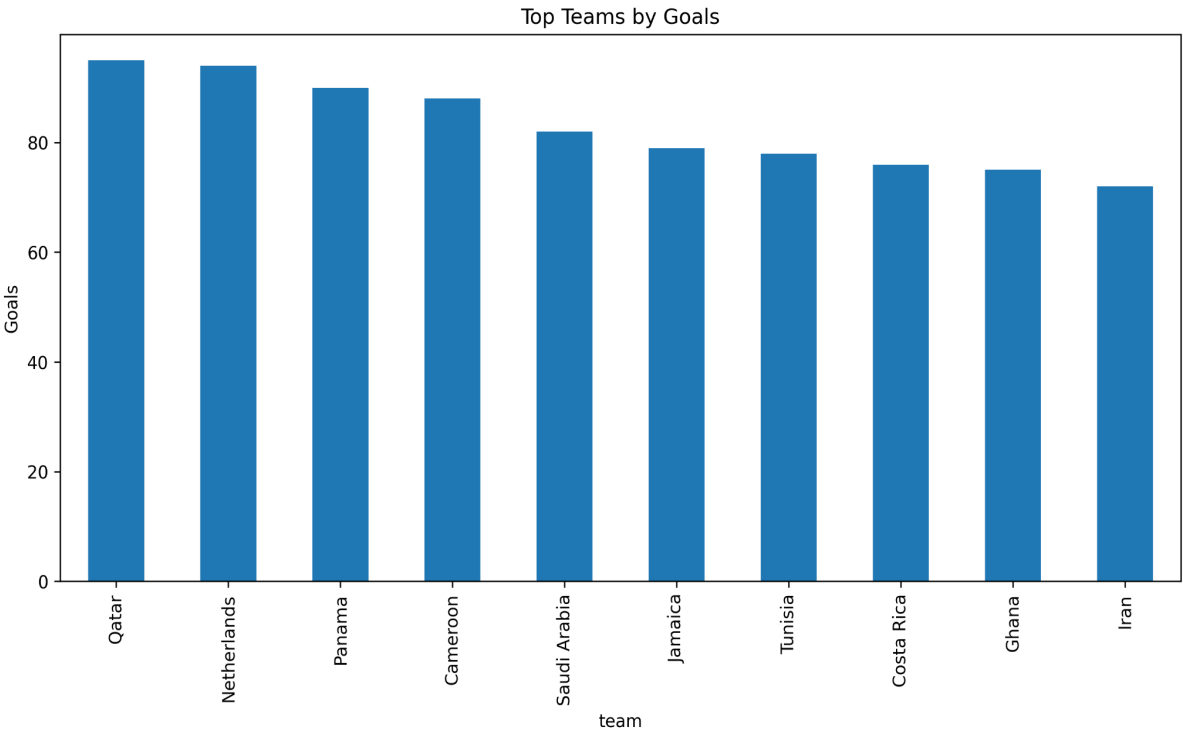
Observation: Memphis Zerrouki emerged as the leading goal scorer. The chart highlights players who made the strongest attacking contributions during the tournament.

Top Assist Providers



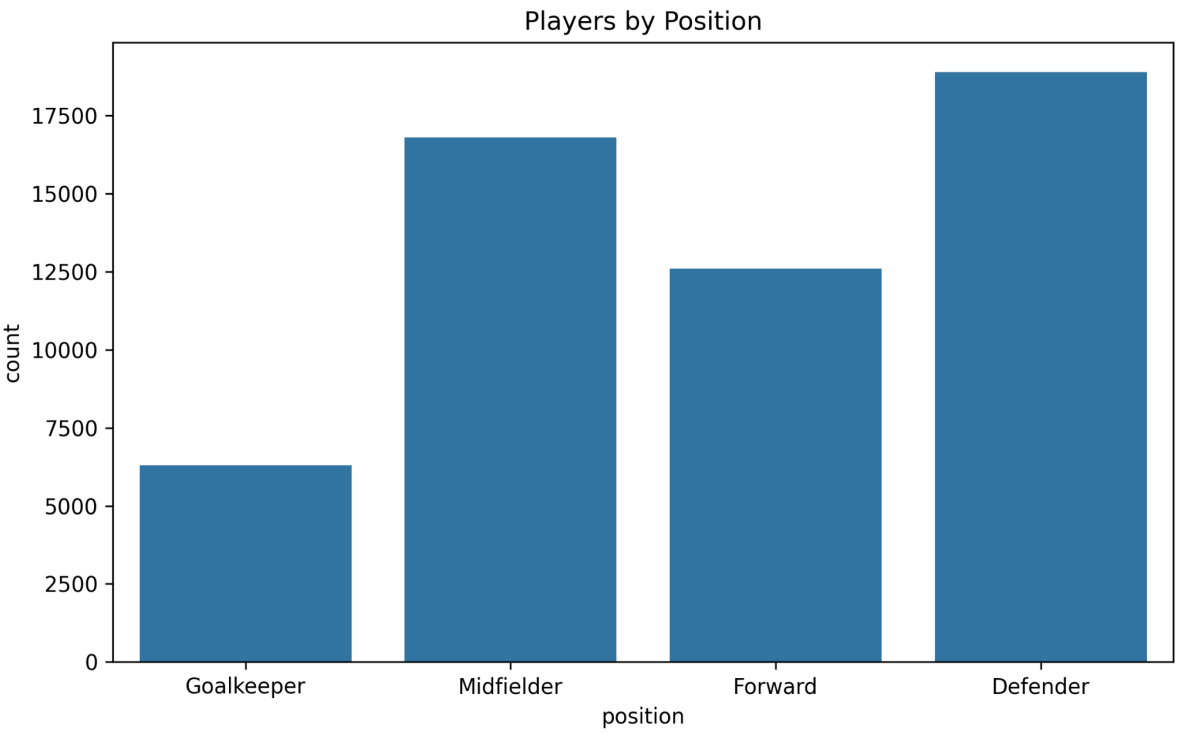
Observation: Yassine El Yamiq recorded the highest number of assists. Assist statistics emphasize creative contributions and playmaking ability.

Team-wise Goal Analysis



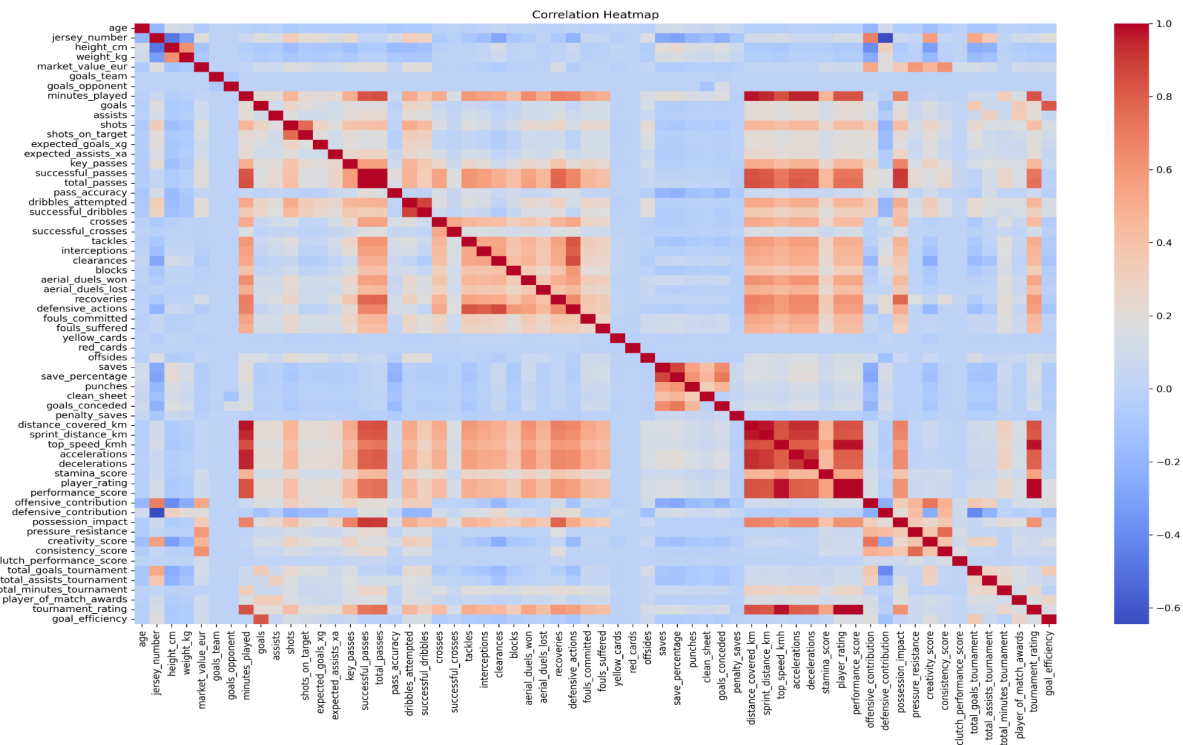
Observation: Qatar and the Netherlands were among the highest-scoring teams. Team scoring trends reveal differences in attacking effectiveness.

Players by Position



Observation: Defenders and midfielders represent the largest share of players in the dataset, while goalkeepers constitute the smallest group.

Correlation Heatmap



Observation: The heatmap reveals strong positive relationships between performance_score, player_rating, tournament_rating, and contribution-based metrics.

Key Insights

1. Mohamed Elneny achieved the highest player rating.
2. Memphis Zerrouki was the top goal scorer.
3. Yassine El Yamiq led the tournament in assists.
4. Defenders formed the largest positional group.
5. Team scoring patterns varied significantly across nations.
6. Player rating showed a strong relationship with performance score.
7. Tournament rating aligned closely with overall player effectiveness.
8. Goal efficiency provides deeper insight into attacking performance.

Challenges Faced

Handling a dataset with numerous variables required careful feature selection and visualization design. Interpreting large correlation matrices and presenting findings clearly were key challenges during the analysis.

Conclusion

This project successfully applied Exploratory Data Analysis techniques to FIFA World Cup 2026 player performance data. The analysis identified top performers, explored team-level trends, examined positional distributions, and revealed relationships among key performance indicators. The project strengthened practical skills in data cleaning, visualization, statistical analysis, and insight generation using Python.

Future Scope

Future work can include machine learning models for player rating prediction, interactive Power BI dashboards, real-time football analytics systems, and advanced performance forecasting techniques.